

KO SAMUI, Thailand

WMO: 485500

Lat: 9.467N

Lon: 100.050E

Elev: 16

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	73.6	74.8	67.2	100.0	77.8	68.8	105.9	79.1	18.6	79.6	16.5	80.6	1.2	30	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.7	92.9	79.9	91.8	79.8	90.7	79.8	82.9	88.7	82.3	88.3	81.6	87.7	5.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	163.3	87.2	80.6	159.0	86.6	79.9	155.4	85.8	46.9	89.1	46.1	88.4	45.4	88.0	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.5	11.2	DB	69.8	95.6	3.5	1.5	67.3	96.6	65.3	97.5	63.3	98.3	60.8	99.4	(4)
(5)			WB	67.0	83.9	3.3	1.6	64.6	85.0	62.7	86.0	60.8	86.9	58.4	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	80.3	81.2	82.7	84.5	85.1	84.6	84.0	84.1	83.3	81.9	80.9	80.4	(6)
(7)		DBStd	2.50	1.69	1.61	1.77	1.98	1.99	2.08	1.94	1.94	1.83	2.00	1.73	1.66	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11955	939	875	1013	1035	1087	1038	1053	1057	998	990	927	943	(10)
(11)		CDD65	6480	474	455	548	585	622	588	588	592	548	525	477	478	(11)
(12)		CDH74	73713	4790	4979	6541	7526	7819	7141	6856	6884	6121	5499	4830	4727	(12)
(13)	CDH80	25579	1016	1293	2312	3315	3461	2972	2629	2630	2111	1602	1184	1054	(13)	
(14)	Wind	WSAvg	3.4	5.1	5.6	4.7	2.9	2.6	2.6	2.7	3.0	2.8	2.3	3.1	3.9	(14)
(15)	Precipitation	PrecAvg	78.0	4.2	2.5	4.3	3.0	5.4	5.0	4.5	4.3	4.6	12.4	18.1	9.8	(15)
(16)		PrecMax	113.1	13.1	15.2	32.9	9.3	12.6	10.0	8.2	14.5	9.1	28.5	42.9	32.0	(16)
(17)		PrecMin	55.2	0.6	0.0	0.0	0.1	1.6	1.5	1.5	0.8	2.3	2.5	5.2	1.6	(17)
(18)		PrecStd	15.7	3.3	3.3	7.1	2.4	2.4	2.2	1.7	2.5	1.8	7.1	9.0	6.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.7	87.8	89.8	93.3	94.4	94.1	93.4	93.6	92.6	91.4	88.2	87.3	(19)
(20)			MCWB	78.3	79.0	79.4	80.6	80.7	80.6	79.5	79.3	79.2	79.3	79.1	78.4	(20)
(21)		2%	DB	85.3	86.0	88.2	90.5	92.4	92.3	91.6	91.8	90.9	89.1	87.0	86.1	(21)
(22)			MCWB	78.2	78.5	79.9	81.0	80.6	80.0	79.3	79.2	78.8	79.3	79.1	78.5	(22)
(23)		5%	DB	84.4	85.1	87.1	89.4	90.9	90.9	90.2	90.5	89.4	87.5	86.1	85.2	(23)
(24)			MCWB	77.8	78.1	79.7	81.1	80.7	79.7	79.2	79.0	78.7	79.3	79.0	78.1	(24)
(25)		10%	DB	83.6	84.3	86.1	88.5	89.7	89.5	88.8	89.0	87.9	86.3	85.2	84.3	(25)
(26)			MCWB	77.5	77.9	79.5	80.8	80.8	79.6	79.1	78.8	78.7	79.1	78.7	77.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.5	80.5	81.9	83.6	84.4	83.1	81.7	81.9	81.6	82.2	81.2	80.9	(27)
(28)			MCDWB	84.1	84.9	86.6	88.7	89.4	90.3	89.2	89.2	87.9	87.5	85.6	84.6	(28)
(29)		2%	WB	79.7	79.8	81.0	82.7	83.3	81.9	80.9	80.8	80.7	80.9	80.5	80.0	(29)
(30)			MCDWB	83.5	84.2	86.0	88.3	88.8	89.0	88.2	88.0	87.2	86.2	85.0	84.0	(30)
(31)		5%	WB	79.1	79.1	80.6	82.0	82.4	81.0	80.2	80.1	80.0	80.1	79.9	79.4	(31)
(32)			MCDWB	83.0	83.7	85.7	87.8	88.4	88.2	87.4	87.2	86.4	85.3	84.3	83.4	(32)
(33)		10%	WB	78.5	78.7	80.0	81.4	81.6	80.3	79.7	79.5	79.3	79.6	79.3	78.8	(33)
(34)			MCDWB	82.6	83.3	85.1	87.3	87.8	87.4	86.7	86.5	85.7	84.6	83.6	82.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	5.7	6.3	7.8	9.7	10.1	10.1	10.2	9.8	8.6	7.5	7.2	(35)
(36)			MCDBR	7.6	7.5	8.1	8.9	11.2	11.8	11.8	12.2	12.1	10.5	9.1	8.4	(36)
(37)			MCWBR	3.8	3.5	3.6	3.6	4.1	4.2	4.2	4.1	4.0	4.0	4.1	4.0	(37)
(38)		5% WB	MCDWB	6.2	6.0	6.8	7.8	9.4	10.7	10.7	10.9	10.4	9.3	7.9	7.4	(38)
(39)			MCWBR	3.6	3.3	3.4	3.7	4.4	4.4	4.6	4.4	4.3	4.2	4.1	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.489	0.497	0.503	0.469	0.443	0.437	0.438	0.431	0.424	0.449	0.472	0.480	(40)	
(41)		taud	2.207	2.179	2.206	2.314	2.437	2.477	2.459	2.466	2.476	2.402	2.310	2.253	(41)	
(42)		Ebn at Noon	261	263	263	269	271	269	270	275	281	274	265	261	(42)	
(43)		Edh at Noon	46	49	48	43	37	35	36	36	36	38	41	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1474	1753	1858	1868	1688	1593	1577	1618	1603	1418	1274	1273	(44)	
(45)		RadStd	137	100	154	119	103	86	107	88	77	129	134	127	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.49	+0.79	N/A	+0.88	+0.50	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.36	+0.63	N/A	+0.65	+0.52	+0.54	N/A	N/A	+133	+131

Nomenclature: See separate page